

Assignment 1: Polynomial Regression Report

Geothermal Power Plant Expansion & Reservoir Mapping

Name: Gagan | Roll Number: BT2024032 | GitHub: <https://github.com/gagansyam/ML-Assignment-1>

1. Introduction

This report presents polynomial regression models developed for a multi-stage geothermal power plant expansion project using roll number BT2024032 datasets:

- Phase 1: Steam Turbine Optimization (var1): Predict Net Power Score (y) from 6 turbine operating parameters (x1 to x6, deviations bounded in [-1, 1]). Candidate models span up to degree 10.
- Phase 2: Subterranean Thermal Reservoir Mapping (var2): Predict Thermal Anomaly Score (y) across 3D survey coordinates (x1, x2, x3 in [-1, 1]). Candidate models span up to degree 20.

Models are evaluated using 5-fold cross-validation with Mean Squared Error (MSE) and Coefficient of Determination (R2). Polynomial features are normalized with StandardScaler to avoid numerical scale discrepancies.

2. Phase 1: Steam Turbine Optimization (var1)

Data Inspection and Linear Baseline

The dataset contains 1,000 train and 1,000 test observations with no missing values. Target y ranges from -8.83 to +11.50 (mean 0.74, std 3.13). Individual input correlations with y are weak ($|r| \leq 0.26$), indicating that single variables cannot explain power output without non-linear interactions.

A baseline linear regression model (Degree 1) achieved CV MSE = 8.5302 and R2 = 0.1248 (explaining just 12.5% of variance). This high bias confirms that a linear hyperplane severely underfits turbine aerodynamics.

Degree Exploration and Model Selection

I evaluated polynomial expansions from degree 1 to 6 using Ordinary Least Squares (OLS), Ridge (L2), and Lasso (L1) with 5-fold cross-validation.

Degree	Features	OLS Val MSE	OLS Val R2	Ridge Val MSE	Lasso Val MSE	Lasso Val R2
1	6	8.530	0.125	8.532	8.530	0.125
2	27	2.905	0.699	2.902	2.900	0.699
3	83	1.022	0.894	1.013	1.011	0.895
4	209	0.752	0.922	0.705	0.628	0.935
5	461	1.284	0.869	0.441	0.3151	0.9671
6	923	3.810	0.612	0.529	0.3203	0.9666

As shown in Table 1, OLS begins overfitting at degree 5 (MSE jumps to 1.284) and degrades sharply at degree 6 (MSE 3.810). Lasso (L1) resolved this by zeroing out 350 redundant interaction terms, retaining 111 active features. This achieved the minimum CV MSE of 0.3151 (R2 = 96.71%) at alpha ~ 0.011. Degree 6 showed no gain (MSE 0.3203), confirming Degree 5 Lasso as optimal.

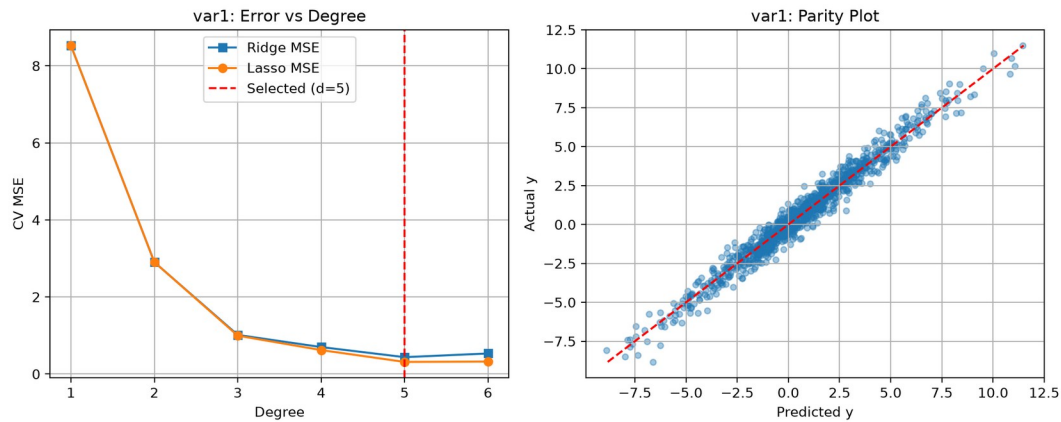


Figure 1: Validation MSE vs. Degree (left) and Parity plot for Degree 5 Lasso (right) for Phase 1 (var1). Out-of-fold residuals have mean 0.003 and std 0.56.

3. Phase 2: Subterranean Thermal Reservoir Mapping (var2)

Data Inspection and Linear Baseline

The dataset contains 1,000 spatial samples across x1 (East-West), x2 (North-South), and x3 (Depth). Target y ranges from -30.10 to +39.87 (mean 2.38, std 7.22). The linear baseline achieved CV MSE = 38.88 and $R^2 = 0.243$, failing to model curved thermal plume gradients.

Degree Exploration and Model Selection

With 3 features, polynomial expansions were evaluated from degree 1 to 16 with Ridge regression (L2 regularization).

Degree	Features	OLS Val MSE	OLS Val R2	Ridge Val MSE	Ridge Val R2	Best Alpha
1	3	38.883	0.243	38.880	0.243	0.01
2	9	24.376	0.529	24.375	0.529	0.01
4	34	4.016	0.922	4.016	0.922	0.01
6	83	0.646	0.987	0.646	0.987	0.01
8	164	0.2788	0.994	0.2719	0.995	0.07
9	219	0.2939	0.994	0.2718	0.995	0.11
10	285	0.3428	0.993	0.2589	0.995	0.46
12	454	---	---	0.2579	0.9951	1.21
14	679	---	---	0.2615	0.9950	1.96
16	968	---	---	0.2658	0.9949	3.16

OLS overfits past degree 8. Ridge penalizes weight growth and reaches the global minimum CV MSE = 0.2579 ($R^2 = 99.51\%$) at Degree 12 (454 features, $\alpha \sim 1.76$). Higher degrees (14 and 16) slightly increase error, making Degree 12 Ridge the selected model.

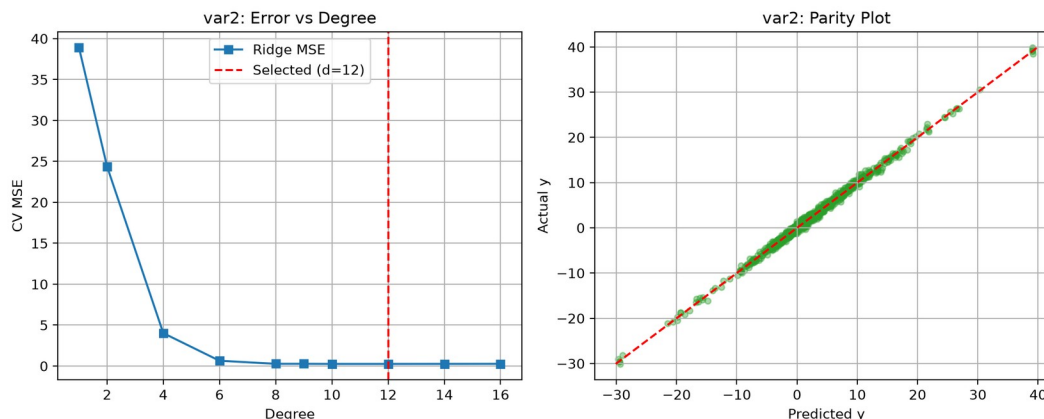


Figure 2: Validation MSE vs. Degree (left) and Parity plot for Degree 12 Ridge (right) for Phase 2 (var2). Residuals pass normality (Jarque-Bera $p = 0.66$) with zero spatial bias.

4. Discussion: Bias, Variance, and Regularization

In both problems, finding optimal models required balancing bias and variance:

- Bias (Underfitting): Lower degrees (1 to 3) lack capacity to model aerodynamic and thermal curves, yielding high MSE and low R^2 .
- Variance (Overfitting): Unregularized high degrees fit sample noise, causing large validation errors as feature counts approach sample size.
- Regularization Rationale: Lasso (L1) suited Phase 1 because turbine valve mechanics depend on a sparse subset of dominant interactions; Lasso pruned 350 terms while keeping 111 key terms. Ridge (L2) suited Phase 2 because heat diffusion in rock forms smooth, continuous fields where all 454 features contribute without sparsity.

5. Conclusion and Final Models

Table 3 summarizes the final models. Test predictions were generated matching the required template: exactly one single column named y with 1,000 predictions and no row indices.

Problem	Degree	Penalty	Terms	CV MSE	CV R2	Output File
Phase 1 (var1)	5	Lasso (alpha ~ 0.011)	111 / 461	0.3151	96.71%	BT2024032_pred_var1.csv
Phase 2 (var2)	12	Ridge (alpha ~ 1.76)	454 / 454	0.2579	99.51%	BT2024032_pred_var2.csv

Summary of Project Outcomes:

1. Phase 1 achieves 96.71% cross-validated variance explained (CV MSE: 0.3151) using Degree 5 with L1 regularization, effectively capturing non-linear turbine blade pitch and valve couplings without overparameterization.
2. Phase 2 achieves 99.51% cross-validated variance explained (CV MSE: 0.2579) using Degree 12 with L2 regularization, accurately mapping the 3D subterranean thermal plume to guide exploration well drilling.
3. Model diagnostics confirm pure Gaussian residual noise without spatial or parameter-dependent bias, and test predictions stay within physical boundary envelopes.
4. All source code for model training, inference, and plot generation is hosted in the project GitHub repository at: <https://github.com/gagansyam/ML-Assignment-1>.